

Assignment05

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
  ↪ "Aquatic_Sampling_Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

Class code

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

```

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),

```

```

        med_sal = median(salinity_ppt, na.rm = TRUE),
        med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality

```

```

left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪   = TRUE),
  site_full = fct_rev(site_full))

```

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median salinity (ppt)",
  title = "Site salinity") +
# different theme
theme_bw()

```

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
    y = "log + 1 transformed average abundance/L",
    title = "Copepods") +
  # different theme
  theme_bw()

```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: different sites
- y-axis: log + 1 transformed average ostracod abundance per liter
- geometry to represent average abundance/L: `geom_jitter()`, circle points
- geometry to represent medians: `stat_summary()`, circle points with a distinct color

b. Wrangle your data

```
# create new object from cleaned aquatic invertebrate data
ostracods_df <- aq_ins_clean |>
  # keep only ostracod rows
  filter(taxon == "ostracod") |>
  # log + 1 transform average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
ostracods_df |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2022-05-13_NVBR	2022-05-13 00:00:00	NVBR	ostr~	0	9.72	74.2	11.4
2	2022-01-21_NEC	2022-01-21 00:00:00	NEC	ostr~	0	NA	NA	NA
3	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.13	NA	NA	NA
4	2022-05-12_NPB	2022-05-12 00:00:00	NPB	ostr~	1.07	7.97	11.4	8.17
5	2023-05-04_NMC	2023-05-04 00:00:00	NMC	ostr~	1.87	8.32	NA	10.3
6	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	ostr~	0.133	7.71	1.98	9.59
7	2024-01-28_NPB	2024-01-28 00:00:00	NPB	ostr~	3.33	7.43	3.08	6.6
8	2023-11-14_NPB	2023-11-14 00:00:00	NPB	ostr~	18	NA	NA	NA
9	2023-05-06_NDC	2023-05-06 00:00:00	NDC	ostr~	56	10.1	NA	13.1
10	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	ostr~	0	NA	NA	NA

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

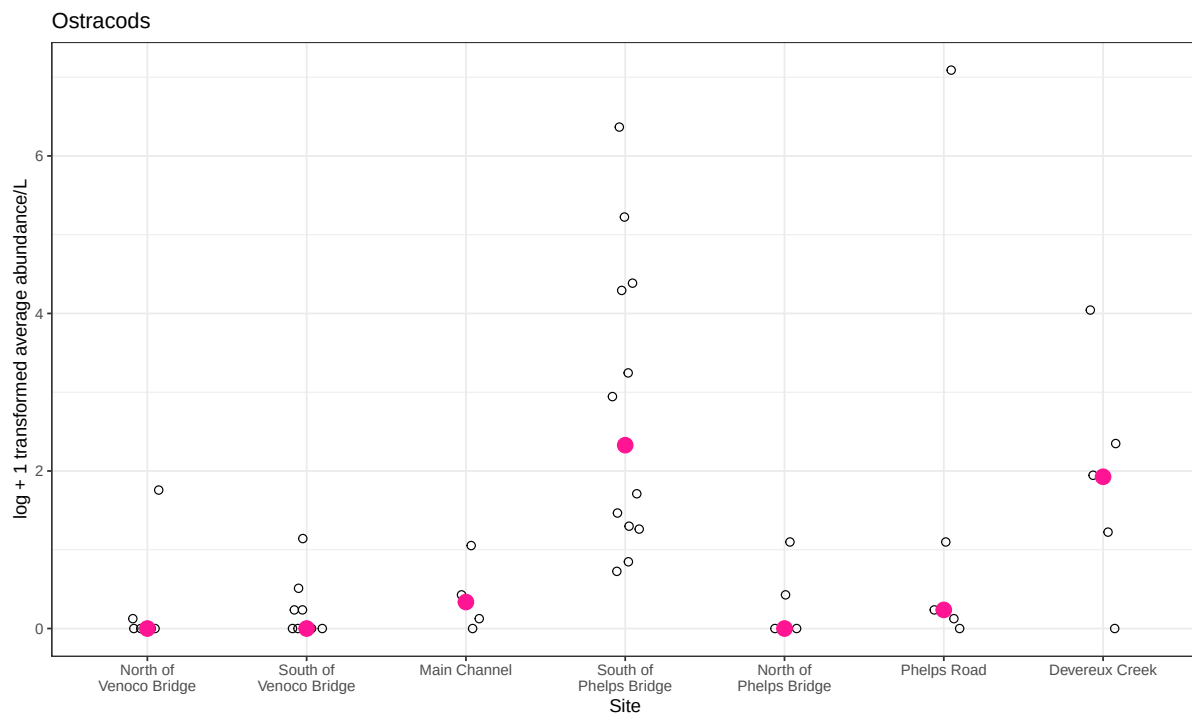
```
# base layer
ostracods <- ggplot(data = ostracods_df,
  # x = site, y = log abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: individual survey points
```

```

geom_jitter(height = 0,
            width = 0.1,
            shape = 21,
            size = 2) +
# second layer: median per site, distinct in size and color
stat_summary(fun = "median",
            geom = "point",
            color = "deeppink",
            size = 4) +
# wrap long site labels for readability
scale_x_discrete(labels = label_wrap(15)) +
# readable axis labels and taxon-specific title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# non-default theme
theme_bw()

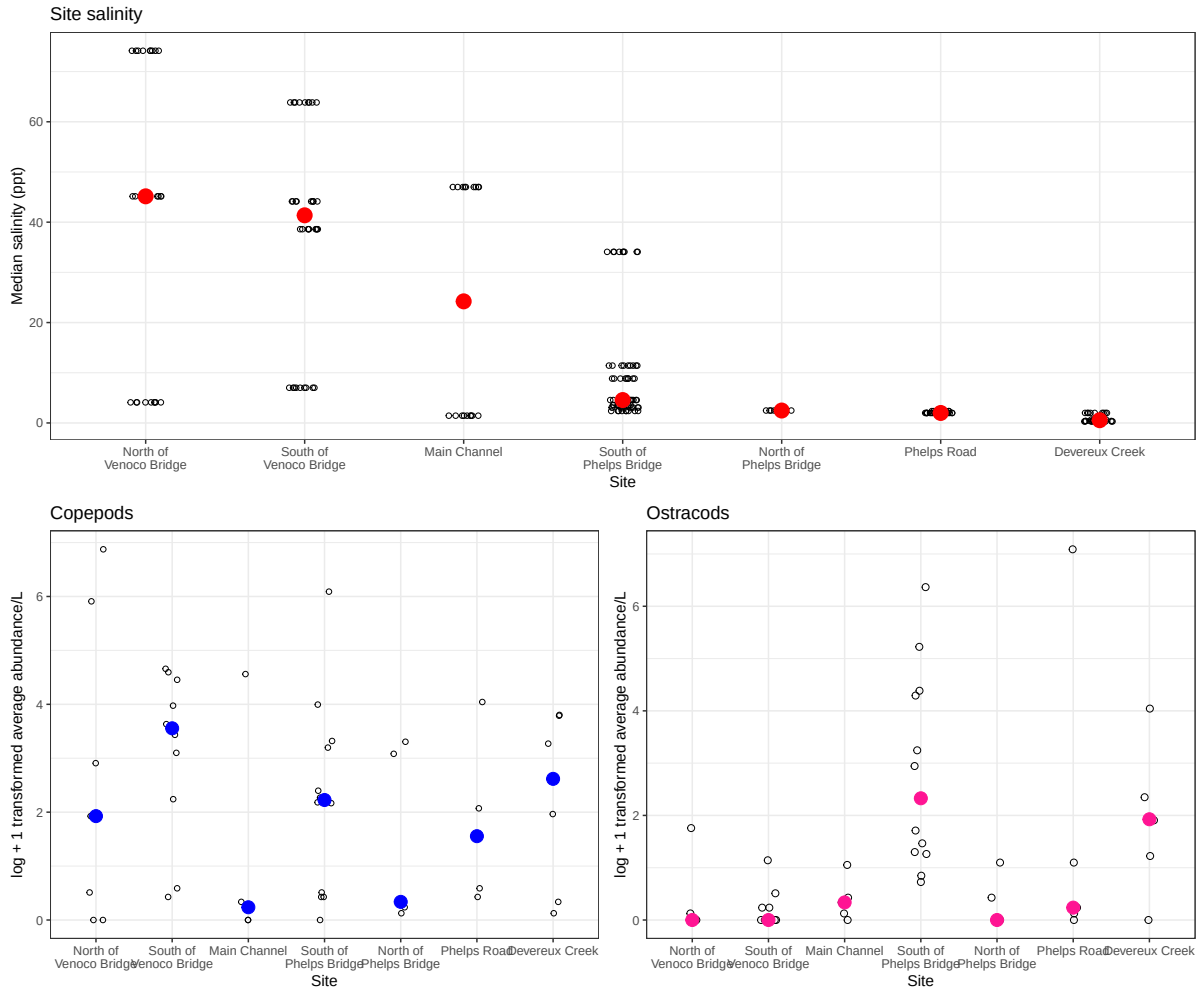
# display object
ostracods

```



d. Create a multipanel figure

```
# salinity on top, copepods and ostracods side-by-side below
(salinity) / (copepods | ostracods)
```



e. Interpret your figure

- Differences between sites in copepod and ostracod abundance: Copepod medians are the blue circles on the bottom left panel. They are generally non-zero at every site. They have largest at South of Venoco Bridge (about 3.5) and the lowest at Main Channel and North of Phelps Bridge (~0.3). Ostracod medians are the pink circles on the bottom right panel. They are essentially zero at the two Venoco Bridge sites and at North of Phelps Bridge. They have sharply risen at South of Phelps Bridge and Devereux Creek.

- How ostracod abundance may relate (or not) to site-level differences in salinity: Because sites are ordered from highest to lowest median salinity left-to-right in the top panel, the pink ostracod medians generally rise as the red salinity medians fall. Ostracods are zero where median salinity exceeds 40 ppt and reach their highest values at sites with median salinity below 5 ppt. The relationship isn't very correlated. Main Channel, North of Phelps Bridge, and Phelps Road all have low ostracod medians despite moderate or low salinity, so salinity alone doesn't fully explain the site-level pattern.

Problem 2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: median salinity, `med_sal`, ppt
- y-axis: log + 1 transformed average abundance per liter, `log_ave_lit`
- color: `taxon`, each taxon own color
- geometry: `geom_point()`, circle points
- facets: different sites

b. Wrangle your data

```
# create new object from cleaned aquatic invertebrate data
abundance_salinity <- aq_ins_clean |>
  # keep only the columns needed for the figure
  select(date_site, site_full, taxon, ave_lit, med_sal) |>
  # log + 1 transform the abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon names to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # order taxon factor levels by maximum abundance
  mutate(taxon = fct_reorder(taxon, ave_lit,
                             .fun = max,
                             .na_rm = TRUE,
                             .desc = TRUE))

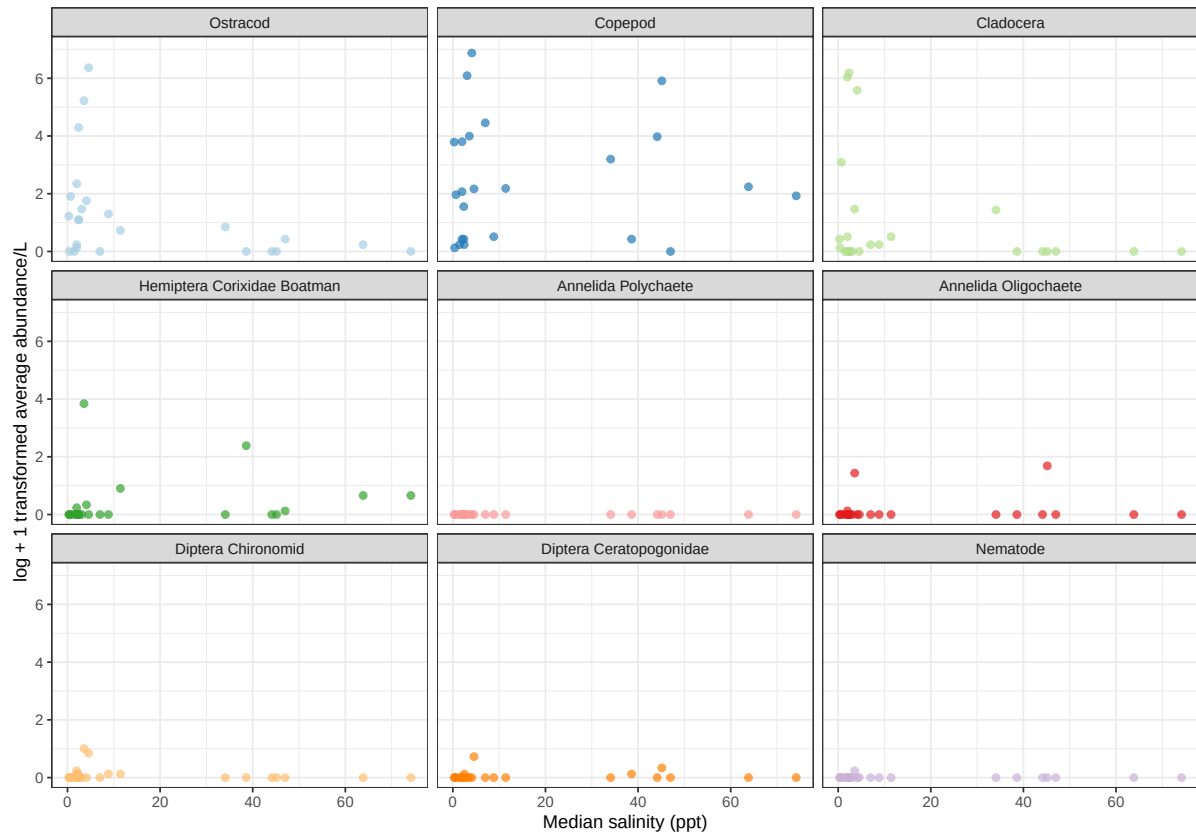
# show 10 random rows
abundance_salinity |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 6
```

	date_site <chr>	site_full <fct>	taxon <fct>	ave_lit <dbl>	med_sal <dbl>	log_ave_lit <dbl>
1	2023-03-03_NDC	Devereux Creek	Diptera C~	0	0.688	0
2	2022-02-25_NEC	South of Venoco Bridge	Diptera C~	0	44.1	0
3	2023-05-04_NMC	Main Channel	Diptera C~	0.267	NA	0.236
4	2022-05-06_NMC	Main Channel	Nematode	0	1.45	0
5	2022-05-12_NPB	South of Phelps Bridge	Ostracod	1.07	11.4	0.726
6	2022-01-24_NEC	South of Venoco Bridge	Hemiptera~	0.267	NA	0.236
7	2022-08-12_NVBR	North of Venoco Bridge	Diptera C~	0	NA	0
8	2023-05-04_NEC	South of Venoco Bridge	Cladocera	0.133	NA	0.125
9	2023-01-28_NPB1	North of Phelps Bridge	Hemiptera~	0	NA	0
10	2022-01-24_NEC	South of Venoco Bridge	Cladocera	0	NA	0

c. Code your figure

```
# base layer
ggplot(data = abundance_salinity,
       # x = salinity, y = log abundance/L, color by taxon
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon)) +
  # points for each survey x taxon observation
  geom_point(alpha = 0.7,
            size = 1.8) +
  # one panel per taxon, free axes
  facet_wrap(~ taxon) +
  # custom palette
  scale_color_brewer(palette = "Paired") +
  # readable axis labels
  labs(x = "Median salinity (ppt)",
       y = "log + 1 transformed average abundance/L") +
  # non-default theme, legend removed
  theme_bw() +
  theme(legend.position = "none")
```



d. Interpret your figure

With fixed y-axes for each panel, the Ostracod, Copepod and Cladocera are relatively large in absolute log-abundance. Each of them reach 6 or above, while the remaining six taxa rarely exceed 2 on the same scale. Ostracod and Cladocera show freshwater bias. Their highest points cluster at salinities below ~10 ppt. And the panels are essentially empty above 20 ppt. Copepod has points reaching above 6 at both freshwater and high-salinity sites. Hemiptera Corixidae Boatman is mostly low-salinity with a relatively low peak near 4 ppt. Annelida Oligochaete, Annelida Polychaete, Diptera Chironomid, Diptera Ceratopogonidae, Nematode are generally near zero.